

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' [11 × 1 = 11] · Time: 25 min

Choose the best alternative.

1. In $F = a\sqrt{x} + bt^2$, where F is force, x is distance and t is time, the dimensions of b/a are
(a) $[L^{1/2} T^{-2}]$ (b) $[M L^{1/2} T^{-2}]$ (c) $[L^{-1/2} T^2]$ (d) $[M L T^{-4}]$
2. Two forces of 6 N and 8 N act at a point. Which of these cannot be their resultant?
(a) 2 N (b) 10 N (c) 14 N (d) 15 N
3. The angle between $A = \hat{i} + \hat{j}$ and $B = \hat{i} - \hat{j}$ is
(a) 0° (b) 45° (c) 90° (d) 180°
4. A car accelerates uniformly from rest to 20 m s^{-1} and then moves uniformly for 5 s. The distance covered in 10 s is
(a) 100 m (b) 150 m (c) 200 m (d) 250 m
5. At the highest point of an oblique projectile, the angle between velocity and acceleration is
(a) 0° (b) 45° (c) 90° (d) 180°
6. A ball is projected at 45° with 20 m s^{-1} ($g = 10 \text{ m s}^{-2}$). Its maximum height is
(a) 5 m (b) 10 m (c) 20 m (d) 40 m
7. On scale X, ice point is 20°X and steam point is 220°X . 50°C on this scale reads
(a) 100°X (b) 110°X (c) 120°X (d) 140°X
8. 100 g of water at 80°C is mixed with 300 g of water at 20°C . The final temperature is
(a) 35°C (b) 40°C (c) 50°C (d) 60°C
9. An object is placed 40 cm in front of a concave mirror of radius of curvature 40 cm. The image is
(a) virtual, erect, magnified (b) real, inverted, same size (c) real, inverted, diminished (d) at infinity
10. A charge q is placed at the centre of a cube. The flux through one face is
(a) q/ϵ_0 (b) $q/6\epsilon_0$ (c) $6q/\epsilon_0$ (d) zero
11. A body acquires a charge of -3.2 uC . The number of electrons it gained is

$$2i \quad \frac{i+j+i-j}{14} \quad \frac{2}{22+10j}$$

MCQ

$$\frac{0}{2}$$

1) $\rightarrow a) L^{\frac{1}{2}} T^{-2}$

2) $\rightarrow a) 15$

3) $\rightarrow a) 0^\circ$

4) $\rightarrow b) 150m$

5) $\rightarrow a) 0^\circ$

6) $\rightarrow b) 10m$

7) $\rightarrow c) 120^\circ X$

8) $\rightarrow a) 35^\circ C$

9) $\rightarrow b) \text{real, inverted same size}$

10) $\rightarrow b) q/6\epsilon_0$

11) $\rightarrow a) 2 \times 10^{13}$

MOCKQ-B

12)

a)

→

The principle of homogeneity states that, the dimensions of the physical quantities in the both sides of the physical equation must be equal.

Eg: $F = ma$
 $[F] = [m][a]$

→ To check correctness of $T = 2\pi\sqrt{\frac{L}{g}}$

$$[T] = [M^0 L^0 T]$$

$$\begin{aligned} [2\pi\sqrt{\frac{L}{g}}] &= [2\pi] \sqrt{\frac{[L]}{[g]}} \\ &= \sqrt{\frac{L}{[g]}} \\ &= \sqrt{\frac{[L]}{[L T^{-2}]}} \\ &= \sqrt{T^2} \\ &= [T] = [M^0 L^0 T] \end{aligned}$$

∴ Correct ✓

Ans

b)
→ Soln,

$$T = \frac{\text{Stress}}{\text{Strain}} = \frac{F}{\frac{\Delta L}{L}}$$

$$\begin{aligned} \text{Dim. } L^1 &= T^{-1} L^{-1} \\ \text{Imp. } [M L^{-1} T^{-2}] &= \frac{[M L^{-1} T^{-2}]}{[L L^{-1}]} \\ &= [M L^{-1} T^{-2}] \end{aligned}$$

$$f \propto L^a T^b U^c$$

$$f = k L^a T^b U^c \quad k \text{ proportionality constant}$$

$$[F] = \left[\frac{1}{L} \right] = [T^{-1}] = [M^0 L^0 T^{-2}]$$

$$\begin{aligned} \text{comp } [k L^a T^b U^c] &= [L]^a [T]^b [U]^c \\ &= [L]^a [M L^{-1} T^{-2}]^b [M L^{-1}]^c \\ &= [M]^{b+c} [L]^{a+b-c} [T]^{-2b} \end{aligned}$$

Comparing with $[M^0 L^0 T^{-2}]$

$$\begin{aligned} b+c &= 0, \quad a+b-c=0 \quad -2b=-1 \\ c &= -\frac{1}{2}, \quad a + \frac{1}{2} = 0 \quad b = \frac{1}{2} \end{aligned}$$

$$a + \frac{1}{2} + \frac{1}{2} = 0 \quad a = -1$$

$$a = -1, \quad b = \frac{1}{2}, \quad c = -\frac{1}{2}$$

$$\begin{aligned} f &= k L^{-1} T^{\frac{1}{2}} U^{-\frac{1}{2}} \\ f &= \frac{k}{L} \sqrt{\frac{T}{U}} \quad \# \end{aligned}$$

$$\begin{aligned} a &= \frac{1 + (-\frac{1}{2})}{2} \\ &= 0 \end{aligned}$$

13)

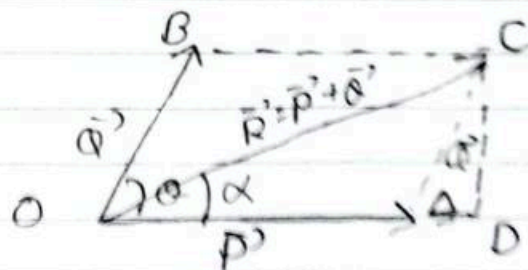
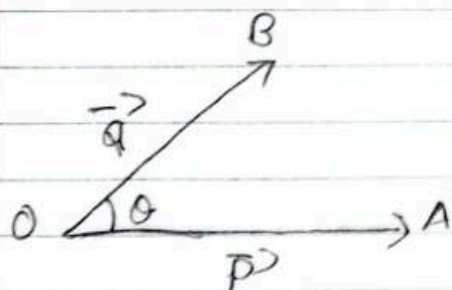
a)

→ Parallelogram law of vector addition states that, "when two vectors act upon a particle at an angle θ , representing its magnitude and direction by the two adjacent sides of a parallelogram, then the diagonal of the parallelogram from that point gives the resultant vector."

Let, P and Q be vectors acting at an angle θ represented by its magnitude and direction by the two adjacent sides OA and OB of the parallelogram $OACD$. then, OC gives the resultant R .

$$\vec{R} = \vec{P} + \vec{Q}$$

$$\vec{OC} = \vec{OA} + \vec{OB}$$



Now, let us draw, $CD \perp OA$ extended to D , then, from right angle triangle OCD

$$\begin{aligned} OC^2 &= \vec{OA}^2 + \vec{AC}^2 \\ &= (\vec{OA} + \vec{AB})^2 + \vec{DC}^2 \quad (1) \end{aligned}$$

Again from, $\triangle ADC$,

$$\sin \theta = \frac{CD}{AC}$$

$$\vec{CD} = AC \sin \theta \quad (ii)$$

$$\cos \theta = \frac{AD}{AC}$$

$$\vec{AD} = AC \cos \theta \quad (iii)$$

Now Putting (ii) and (iii) in eq (i)

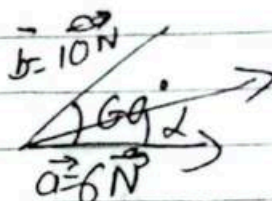
$$\begin{aligned} Nc \quad R^2 &= (\vec{P} + \vec{A} \cos \theta)^2 + AC^2 \sin^2 \theta \\ &= P^2 + 2PQ \cos \theta + Q^2 \cos^2 \theta + Q^2 \sin^2 \theta \\ &= P^2 + 2PQ \cos \theta + Q^2 (\sin^2 \theta + \cos^2 \theta) \\ |\vec{R}| &= \sqrt{P^2 + 2PQ \cos \theta + Q^2} \end{aligned}$$

For direction, In $\triangle ODC$

$$\begin{aligned} \phi \quad \tan \phi &= \frac{CD}{OD} \\ &= \frac{AC \sin \theta}{OA + AD} \\ &= \frac{Q \sin \theta}{P + Q \cos \theta} \end{aligned}$$

b1
→ Given,

$$\vec{a} = 6\text{ N}$$
$$\vec{b} = 10\text{ N}$$



$$R = \sqrt{a^2 + b^2 + 2ab \cos 60}$$
$$= \sqrt{6^2 + 10^2 + 2 \times 6 \times 10 \cos 60}$$
$$= 14\text{ N}$$

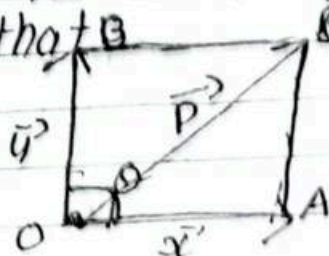
$$\text{Dir, } \alpha = \tan^{-1} \left(\frac{a \sin \theta}{b + a \cos \theta} \right)$$
$$= \tan^{-1} \left(\frac{6 \sin(60)}{10 + 6 \cos(60)} \right)$$
$$= 21.7867^\circ \text{ from } \vec{a}$$

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a'
→

Resolution of a vector is the process of splitting a vector into two or more vectors, which if combined gives the original vector.

→ Resolving a vector into two rectangular components:

let, $\vec{P}$ be the vector, and, a rectangle $OABC$ is formed joining the line, $OC(\vec{P})$ such that $OB \perp OA$, $\theta = 90^\circ$



Now, let the OA' be $\vec{x}$ and OB' be $\vec{y}$, then, from right angled triangle, OAC ,

~~$$\sin \theta = \frac{AC}{OC}$$~~ since, $AC = OB$

~~$$\sin \theta = \frac{OB}{OC}$$~~

$$OC \sin \theta = OB$$

$$|\vec{P}| \sin \theta = |\vec{y}|$$

$$y = p \sin \theta, \quad (i)$$

$$\cos \theta = \frac{OA}{OC}$$

$$\cos \theta = \frac{|\vec{x}|}{|\vec{P}|}$$

$$p \cos \theta = x$$

$$x = p \cos \theta \quad (ii)$$

From (i) and (ii)

vertical component = $y = p \sin \theta$

horizontal component = $p \cos \theta$

Q.E.D.

$$A = 2 \quad \sqrt{2^2 + 3^2}$$

$$B = 3 \quad =$$

b1
→ Soln. Given,

$$\vec{A} = 2\hat{i} + 3\hat{j} - k$$

$$\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$$

$$\theta = ?$$

We know,

$$A \cdot B = AB \cos \theta$$

$$\sin \theta = \frac{|\vec{A} \times \vec{B}|}{AB}$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\theta = \sin^{-1} \left(\frac{|\vec{A} \times \vec{B}|}{AB} \right)$$

$$|\vec{A} \times \vec{B}| = (2\hat{i} + 3\hat{j} - k) \times (\hat{i} - 2\hat{j} + 4\hat{k})$$

$$= (2 - 6 - 4)$$

$$= (-8)$$

$$AB = \sqrt{2^2 + 3^2 + 1^2} \times \sqrt{1^2 + 2^2 + 4^2}$$

$$= \sqrt{14} \times \sqrt{21}$$

$$= 7\sqrt{6}$$

$$\theta = \sin^{-1} \left(\frac{-8}{7\sqrt{6}} \right) \quad \theta = \cos^{-1} \left(\frac{-8}{7\sqrt{6}} \right)$$

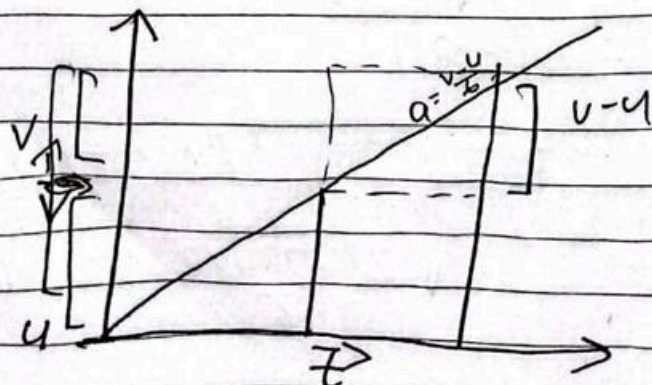
$$= 117.81184^\circ$$

c)
 → No a rectangular component of a vector cannot be greater than the vector itself either it can be smaller or same, because for eg, horizontal component = $|\vec{a}| \cos \theta$
 here, $|\vec{a}| \cos \theta \leq |\vec{a}|$,

15)

a) ~~1st~~

→ Soln, a velocity time graph is drawn,



We know, $s = \frac{v+u}{2} \cdot t$ $s = \frac{v}{2} \cdot t$

So the area under the graph gives

$$\frac{b}{(n-1) \times (n-1)} \frac{1}{t} =$$

$$7 \cdot (n-1) \frac{1}{t} =$$

$$\left(\frac{b}{n-1} \right) \frac{1}{(n-1)} \frac{1}{t} =$$

$$7 \cdot (n-1) \frac{1}{t} = 5$$

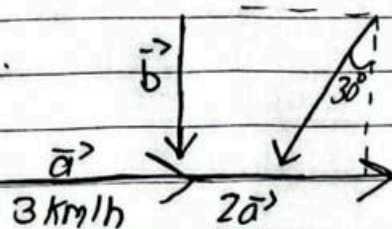
the distance,

b) From eq (1)

$$\vec{a} = 3\text{km}$$

$$b = ?$$

$$\theta = 90^\circ$$



a. $\vec{a} = 3\hat{i}$
 $\vec{b} = -4\hat{j}$

$$\vec{a} + \vec{b} = 3\hat{i} - 4\hat{j} \quad (i)$$

Case (ii)

$$\vec{a} = 6\hat{i}$$

$$\vec{b} = -b \sin 30^\circ \hat{i} + b \cos 30^\circ \hat{j}$$

$$\begin{aligned} \vec{a} + \vec{b} &= 6\hat{i} - b \sin 30^\circ \hat{i} + b \cos 30^\circ \hat{j} \\ &= (6 - b \sin 30^\circ) \hat{i} + b \cos 30^\circ \hat{j} \end{aligned}$$

$$3\hat{i} - 4\hat{j} = (6 - b \sin 30^\circ) \hat{i} + b \cos 30^\circ \hat{j}$$

$$R = \sqrt{3^2 + 4^2}$$

$$R = \sqrt{(6 - b \cos 30^\circ)^2 + (b \sin 30^\circ)^2}$$

$$25 = 1$$

$$25 = (6 - b \cos 30^\circ)^2 + (b \sin 30^\circ)^2$$

$$25 = 36 - 2 \cdot 6 \cdot b \cos 30^\circ + b^2 \cos^2 30^\circ + b^2 \sin^2 30^\circ$$

$$25 = 36 - 12b \cos 30^\circ + b^2$$

$$b^2 - 12b \cos 30^\circ = -11$$

$$b^2 - 12b \cdot \frac{\sqrt{3}}{2} + 11 = 0$$

$$b^2 - 6\sqrt{3}b + 11 = 0$$

$$|b| = 0.19615 \text{ m/s km/hr}$$

16)

a)

→

$$\frac{x - L.F.P_x}{H.B.P_x - L.F.P_x} = \frac{y - L.F.P_y}{H.B.P_y - L.F.P_y}$$

When, x is temp in one scale and y is temp in another scale.

L.F.P = low Freezing point
~~H.B.P~~ = High Boiling point

$$\frac{x - L.F.P_x}{U.F.P_x - L.F.P_x} = \frac{y - L.F.P_y}{U.F.P_y - L.F.P_y}$$

b)

Given, For Z,

$$U.F.P_z = 180^\circ Z$$

$$L.F.P_z = -20^\circ Z$$

For Celsius,

$$U.F.P = 100^\circ C$$

$$L.F.P = 0^\circ C$$

GASE I,

$$x = y$$

$$\frac{x - U.F.P_z}{U.F.P_z - L.F.P_z} = \frac{x - L.F.P}{U.F.P - L.F.P}$$

$$x + 20 = x - 0$$

$$180 + 20 = 100 - 0$$

$$100(x + 20) = x \times (200)$$

$$100x + 2000 = 200x$$

$$2x = x + 20$$

$$x = 20 \quad \text{At } 20^\circ Z$$

2 reading and 1 reading will be same

CASE 2 :

$$x = ?$$

$$y = 40^\circ$$

$$\frac{x+20}{180+20} = \frac{210}{100}$$

$$x+20 = \frac{y}{10} \times 200$$

$$x = 80 - 20 \\ = 60^\circ \#$$

c) $\rightarrow$ Soln, $\Delta\theta = \theta_2 - \theta_1$
 $= 40 - 20$
 $= 20^\circ\text{C}$ (As temp is increased it will lose time)

We know, $T_{\text{loss/day}} = \frac{1}{2} \alpha \Delta\theta \times 86400$
 $= \frac{1}{2} \times 1.9 \times 10^{-5} \times 20 \times 86400$
 $= 16.416 \text{ sec}$

It will lose 16.416 sec per day at 40°C //

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- a) of a body
 → Water equivalent refers to the amount of water needed to absorb the same amount of energy^{heat} that a body has gained or lost,

$$W_{\text{equiv.}} = ms \, dt$$

$$W \times s_w \, dt = ms \, dt$$

$$W = \frac{ms}{s_w}$$

$$= \frac{C}{s_w}$$

$$= \frac{C}{s_w}$$

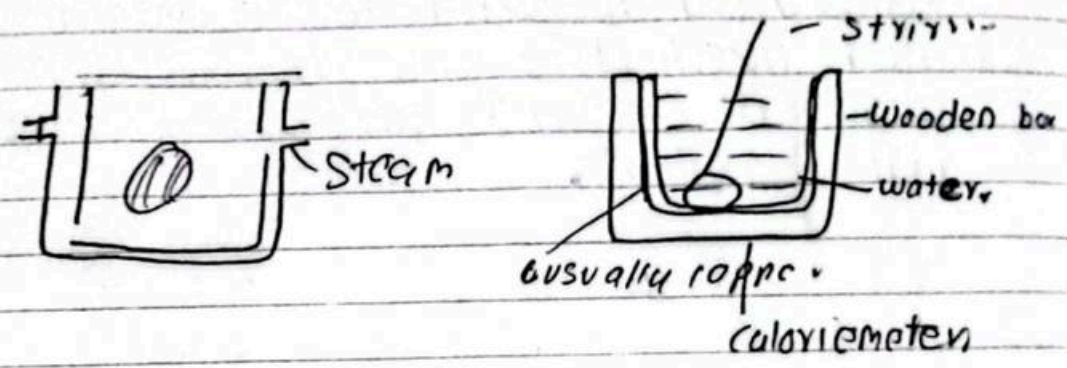
$$W = \frac{C}{1000} \quad \text{In S.I unit,}$$

- It is different from heat capacity because heat capacity is the amount of heat needed to raise the temp of any mass of a substance by 1°C ,

$$C = m \cdot s$$

b)
→

DETERMINATION OF S.P. HEAT CAPACITY OF A SOLID BY THE METHOD OF MIXTURES.



- 1) The substance whose S.P.C is to be measured is, heated up to certain temperature.
- 2) Then the substance is transferred to the calorimeter, and stirred until it reaches the final temp.
- 3) Then, the final temp of the mixture is noted.

Heat lost by substance = Heat gained by calorimeter and water.

let, m_s (mass of substance) = m_s
 (mass of calorimeter) = m_c
 (mass of water) = m_w
 S.P. of substance = S_s
 S.P. of calorimeter = S_c
 S.P. of water = S_w

Initial temp of substance $\doteq \theta_s$

Initial temp of water and calorimeter $\doteq \theta_{cw}$

Final temp of mixture $= \theta$

Heat lost by substance = Heat gained by
calorimeter
and water,

$$m_s s_s (\theta_s - \theta) = m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})$$

$$s_s (\theta_s - \theta) = \frac{m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})}{m_s}$$

$$s_s = \frac{m_c s_c (\theta - \theta_{cw}) + m_w s_w (\theta - \theta_{cw})}{m_s (\theta_s - \theta)}$$

$$s_s = \frac{(\theta - \theta_{cw}) [m_c s_c + m_w s_w]}{m_s (\theta_s - \theta)}$$

c)

→ Given,

 m m Mass of metal (m_m) = 0.2 kgInitial temp of metal (θ_m) = 100°CMass of water (m_w) = 0.15 kg

Mass of

Water equivalent of calorimeter

(c_c) = 0.01 kgor, $m_c s_c = 0.01$ final temp (θ) = 30°C $s_w = 4200 \text{ J/kg}^\circ\text{K}^{-1}$

Heat lost by metal = Heat gained by water + calorimeter

$$m_m s_m (\theta_m - \theta) = m_c s_c (\theta - 20) + m_w s_w (\theta - 20)$$

$$s_m = \frac{m_c s_c (\theta - 20) + m_w s_w (\theta - 20)}{m_m (\theta_m - \theta)}$$

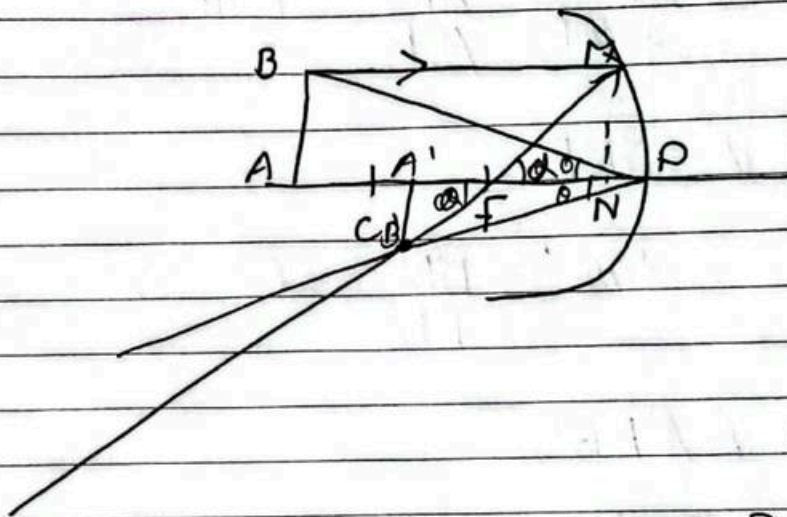
$$= \frac{0.01 (30 - 20) + 0.15 \times 4200 (30 - 20)}{0.2 (100 - 30)}$$

$$= 450.0071 \text{ J/kg}^\circ\text{K}^{-1}$$

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a)

Deriving mirror formula for a concave mirror forming a real image.



Let us consider a concave mirror of focal length f , center of curvature C and pole P , then, an object AB is placed behind C , then real and inverted image $A'B'$ is formed between f and C ,

$$\text{let, } u = AP$$

$$v = A'P$$

$$f = FP$$

Now, for $\triangle APB$

$$\tan \theta = \frac{AB}{AP} \quad \text{--- (i)}$$

In $\triangle A'B'P$

$$\tan \theta = \frac{A'B'}{A'P} \quad \text{--- (ii)}$$

From (i) and (ii)

$$\frac{AB}{AP} = \frac{A'B'}{A'DP}$$

$$\frac{AB}{A'B'} = \frac{AP}{A'P} \quad \text{--- (iii)}$$

In $\triangle MNE$,

$$\tan \alpha = \frac{MN}{FN} \quad \text{--- (iv)}$$

In $\triangle A'FB'$,

$$\tan \alpha = \frac{A'F}{A'B'} \quad \text{--- (v)}$$

From, (iv) and (v)

$$\frac{MN}{FN} = \frac{A'B'}{A'F}$$

or; $\frac{MN}{A'B'} = \frac{FN}{A'F} \quad \text{--- (vi)}$

Since, the aperture is very small,

$$\left. \begin{array}{l} FM = FP \\ MN = AB \\ FN = RP \end{array} \right\} \text{--- (vii)}$$

Putting (vii) in eq (vi)

$$\frac{AB}{A'B'} = \frac{FP}{A'F} \quad \text{--- (viii)}$$

From, eq (viii) and (ii)

$$\frac{AB}{A'B'} = \frac{AP}{A'P} = \frac{FP}{A'F}$$

$$\text{or, } \frac{u}{v} = \frac{f}{u-f}$$

$$uv - uf = vf$$

$$\frac{uv}{uvf} = \frac{uf}{uvf} = \frac{vf}{uvf}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad \#$$

CONVENTIONS:

- i) The aperture is very small
- ii) The ~~ima~~ ray passes thru f ,
- iii) Real image is formed,
- iv) Obj is behind $2f$,

b)

→ Given,

$$\text{size of image} = (S_i)$$

$$\text{size of obj} = (S_o)$$

$$u = 12 \text{ cm}$$

$$f = ?$$

Given,

$$S_i = 4 S_o$$

We know,

$$m = 4$$

$$\frac{S_i}{S_o} = \frac{v}{u} = 4$$

$$\frac{4 S_o}{S_o} = \frac{v}{u}$$

$$4 = \frac{v}{u}$$

$$a. \quad 12 \times 4 = v$$

$$40 + 8 = v$$

$$b. \quad v = 48 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$= \frac{1}{12} + \frac{1}{48}$$

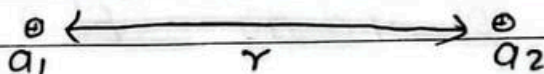
$$\frac{1}{f} = \frac{5}{48}$$

$$f = \frac{48}{5} = 9.6 \text{ cm}$$

10)

a)

→ Coulomb's law states that, "the total force of attraction or repulsion is ~~div~~ between any two point charges is ~~div~~ inverse (~~the~~ proportional) to the distance between them and the product of the two charges." directly proportional



From Coulomb's law,

$$F \propto \frac{K q_1 q_2}{r^2}$$

$$F = K \frac{q_1 q_2}{r^2}$$

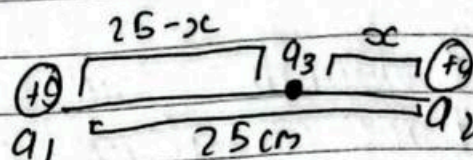
$$\frac{q^2}{r^2} K = F \quad q^2 = \frac{F r^2}{K}$$

→ ~~1 coul~~

1 coulomb of charge is defined as the charge of two point charges required to produce a force of attraction or repulsion that, puts them 1m apart, in vacuum or air.

b1

→ Given,



Given,

$$q_1 = 9 \times 10^{-6} \text{ C}$$

$$q_2 = 4 \times 10^{-6} \text{ C}$$

$$r = 25 \text{ cm}$$

$$x = ?$$

Force Electrostatic force produced betw
een, q_1 and q_3 $(E_{13}) = \frac{k q_1 q_3}{(25-x)^2}$

Electrostatic force produced b-
etw q_3 and q_2 ;

$$E_{32} = \frac{k q_2 q_3}{x^2}$$

$$E_{13} = E_{32}$$

$$\frac{k q_1 q_3}{(25-x)^2} = \frac{k q_2 q_3}{x^2}$$

$$\frac{9 \times 10^{-6}}{(25-x)^2} = \frac{4 \times 10^{-6}}{x^2}$$

$$\left(\frac{x}{25-x} \right)^2 = \frac{4 \times 10^{-6}}{9 \times 10^{-6}}$$

$$\frac{x}{25-x} = \sqrt{\frac{4}{9}}$$

$$5x = 50$$

$$x = 10 \text{ m}$$

$$\frac{x}{25-x} = \frac{2}{3}$$

$$3x = 50 - 2x$$

~~The third charge~~ The point where the net force on 3rd charge is 0 is at 10 m from (+4 μ C).

20)

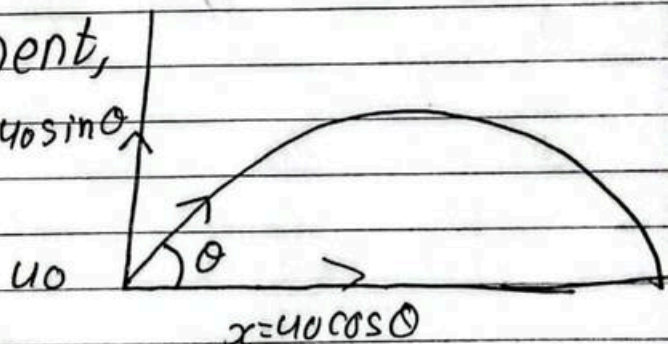
a)

→ Suppose a body is projected from the ground with velocity u at an angle θ with the horizontal, then u can be result into two rectangular components.

For ~~re~~ horizontal component,

velocity $= u \cos \theta$ $u_y = u \sin \theta$

For vertical,



velocity $= u \sin \theta$ decreases

(Note: this velocity increases as it moves towards top and increases as it moves down,)

The eq of path of the projectile,
let $P(x, y)$ be the position after time t .
The projectile travels, x (horizontal) distance and y vertical distance as a function of time.

For horizontal distance.

$$s_x = u_x t + \frac{1}{2} a_x t^2$$

$$s_x = x, \quad u_x = u \cos \theta, \quad a_x = 0$$

$$x = u \cos \theta t$$

$$t = \frac{x}{u \cos \theta} \quad \text{--- (i)}$$

for vertical distance.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = u \sin \theta t - \frac{1}{2} g t^2$$

$$= u \sin \theta \cdot \frac{x}{u \cos \theta} - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

$$y = \tan \theta x - \left(\frac{g}{2u^2 \cos^2 \theta} \right) x^2$$

The path of the projectile, is a parabola as, it takes the form of,

$$y = Ax + Bx^2 + C$$

Time of flight :

The total time it takes for the projectile to reach the top height and come down to the ground is called time of flight.

Since, $v = 0 \text{ m/s}$, when the projectile reaches the top most height.

$$v = u + at$$

$$0 = u \sin \theta - g t$$

$$t = \frac{u \sin \theta}{g}$$

Since the path of the projectile is symmetrical, the amount of time taken to reach top = amount of time taken to reach ground.

$$T = 2t$$

$$= \frac{2u \sin \theta}{g}$$

MAX height,

When the projectile reaches max height, $v = 0 \text{ m/s}$,

$$v^2 = u^2 + 2as$$

$$0 = u^2 \sin^2 \theta - 2gH$$

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

HORIZONTAL RANGE:

~~The maximum~~ horizontal range takes total time (T) to reach or Time of flight s_o ,

$$\begin{aligned} R &= u \cos \theta \cdot T \\ &= u \cos \theta \cdot 2u \sin \theta \\ &= \frac{u^2 \sin 2\theta}{g} \end{aligned}$$

b)

→ let us suppose, x be the range to be covered,

then, θ_1 be the 1st angle and θ_2 be the 2nd angle,

$$\begin{aligned} \text{let, } x &= 50\text{m} & t &= 10\text{sec} \\ \theta_1 &= 30^\circ \\ \theta_2 &= 60^\circ \end{aligned}$$

$$u_{01} = ?$$

$$u_{02} = ?$$

Now,

$$x = \frac{u \cos \theta \cdot t}{\cos \theta}$$

$$u_{01} = \frac{x}{t}$$

$$= \frac{50 \cdot 10}{\cos 30^\circ} = 577.35\text{m/s}$$

$$u_{02} = \frac{50 \times 10}{\cos(60^\circ)} = 1000 \text{ m/s}$$

Now, when $\theta = 30^\circ$

$$\begin{aligned} x &= \frac{u_0 \cos \theta}{t} \\ &= \frac{577.35 \times \cos(30^\circ)}{10} \\ &= 50 \text{ m} \end{aligned}$$

When, $\theta = 60^\circ$

$$\begin{aligned} x &= \frac{u_0 \cos \theta}{t} \\ &= \frac{1000 \times \cos(60^\circ)}{10} \\ &= 50 \text{ m} \end{aligned}$$

$\therefore$ Hence, two different angles can obtain same horizontal range.

c)

→ Given,

$$u_0 = 25 \text{ m/s}$$

$$\theta = 53^\circ$$

i) Time of flight

$$T = \frac{2u_0 \sin \theta}{g}$$

$$= \frac{2 \times 25 \times \sin(53^\circ)}{10}$$

$$= 3.99 \text{ sec}$$

ii) MAX height:

$$H = \frac{u_0^2 \sin^2 \theta}{2g}$$

$$= \frac{25^2 \times (\sin 53^\circ)^2}{2 \times 10}$$

$$= 19.93 \text{ m}$$

iii) Max range,

$$R_{\max} = \frac{u_0^2 \sin 2\theta}{g}$$

$$= \frac{25^2 \times \sin(2 \times 53^\circ)}{10}$$

$$= 23.8 \text{ m}$$

$$\frac{u_0^2 \sin 2\theta}{g}$$

$$= \frac{25^2 \times \sin(2 \times 53^\circ)}{10}$$

$$= 23.8 \text{ m}$$

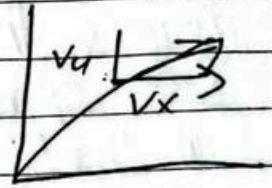
$$= \frac{25^2 \times \sin(2 \times 53^\circ)}{10}$$

$$= 23.8 \text{ m}$$

V after 1 sec,

$$t = 1 \text{ sec}$$

$$V = \sqrt{V_x^2 + V_y^2}$$



$$V_x = u \cos \theta$$

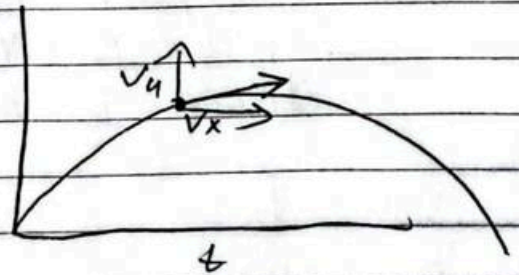
$$V_y = ?$$

$$V_y = u + at$$

$$V_y = u \sin \theta - gt$$

$$\begin{aligned} V &= \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2} \\ &= \sqrt{(25 \times \cos(57))^2 + (25 \times \sin(57) - 10)^2} \\ &= 17.48 \text{ m/s,} \end{aligned}$$

2)



2/1

a)

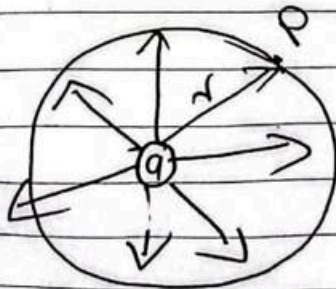
→

Gauss theorem states that, "In any enclosed space the electric flux passing through that space in vacuum is $\frac{1}{\epsilon_0}$ times the total charge enclosed by that space."

$$\Phi = \frac{q}{\epsilon_0}$$

Proving $\Phi = \frac{q}{\epsilon_0}$,

Let, ~~P be the~~ ^{q be} point charge of radius r ,
and, P be the point at the distance R , in which a gaussian sphere is drawn so that it encloses the q .



We know, $\Phi = \sum A \cos \theta$
 $\Phi = \sum A$ [$\theta = 0$ as, $\perp$]
 $= \epsilon \times 4\pi r^2$

We know,

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

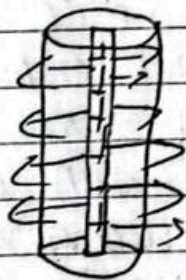
$$\Phi = \frac{q}{4\pi\epsilon_0 r^2} \cdot 4\pi r^2$$

$$\Phi = \frac{q}{\epsilon_0} \quad \text{Proved} \quad \#$$

b)

→ Let, the length of infinitely long straight uniformly charged wire, be L

the, Surface charge density,
 $\lambda = \frac{q}{L}$



Now, let us draw a cylindrical gaussian surface at a distance r from the rod, enclosing it,

$$\text{the, } \Phi_{\text{curved}} = EA = E \times 2\pi r l$$

$$\Phi_{\text{cross section}} = EA \cos(90^\circ) = 0$$

from,

$$E = \frac{F}{q}$$

$$\Phi = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$

$$\epsilon_0 E 2\pi r l = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{2\pi r l \epsilon_0}$$

$$E = \frac{q}{2\pi r \epsilon_0}$$

1)
→

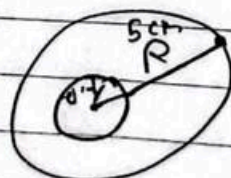
Two electric field lines never intersect each other because, the intersection, of two electric field lines tangent drawn at the intersection gives us two direction, which violates vector law.

2)
→

Given, $r = 0.1 \text{ m}$
 $q = 2 \times 10^{-8} \text{ C}$

i) E at, 5 cm from it's center,

$$E = \frac{q}{4\pi r^2 \epsilon_0}$$



Here $r = 0.1 + 0.05 = 0.15 \text{ m}$

$$\begin{aligned}
 \epsilon &= \frac{2 \times 10^{-8}}{4\pi r^2 \epsilon_0} \\
 &= \frac{9 \times 10^9 \times 2 \times 10^{-8}}{(0.15)^2} \\
 &= 8000 \text{ N/C}
 \end{aligned}$$

ii) ϵ at 20 cm from its center.

$$\begin{aligned}
 r &= 0.1 + 0.2 \text{ m} \\
 &= 0.3 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \epsilon &= \frac{9 \times 10^9 \times 2 \times 10^{-8}}{(0.3)^2} \\
 &= 2000 \text{ N/C}
 \end{aligned}$$

22)

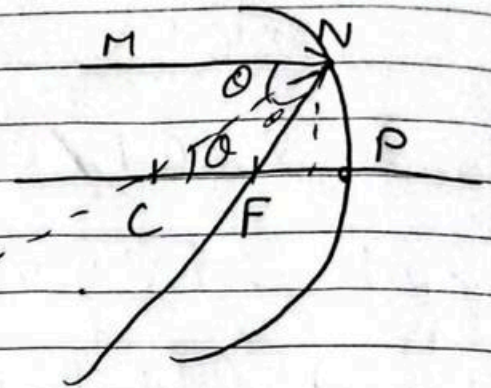
a)

Radius of curvature is defined as the radius of the original sphere from which the spherical mirror was derived.

b) Focal length of a spherical mirror is defined as the length of focus from pole of the spherical mirror.

R between C and F

let us suppose a
concave mirror of
focus F and center
of curvature C,
a ray MN
parallel to the principal
axis, the passing
thru, F



$$\text{let, } R = CP \\ F = FP$$

Now, ~~in $\triangle MNO$, $\angle$~~

$$\angle MNO = \angle ONF \quad (\text{ik } \angle i = \angle r) \\ \angle MNC = \angle NCF \quad (\text{ii}) \quad (\text{Alternate})$$

from, (i) and (ii)

$\angle ONF = \angle NCF$,
as, two angles are equal $\triangle NCF$ is
an isosceles triangle,

$$\therefore CF = FN$$

Using convention, aperture is very
small,

$$FN = FP$$

$$CF = FP$$

$$\begin{aligned} \text{No. } R &= CP \\ &= (F + FP) \\ &= FP + FP \\ &= f + f \end{aligned}$$

$$R = 2f \quad \square$$

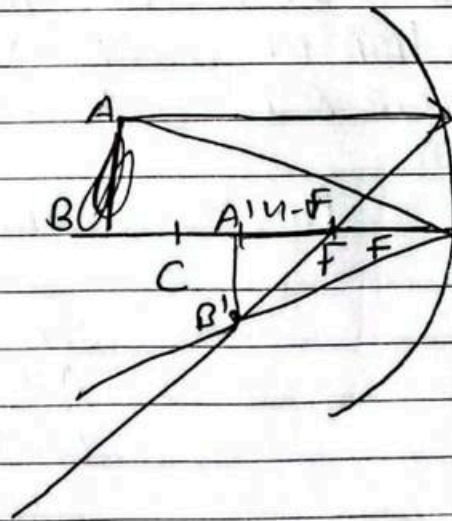
(b)

Now,

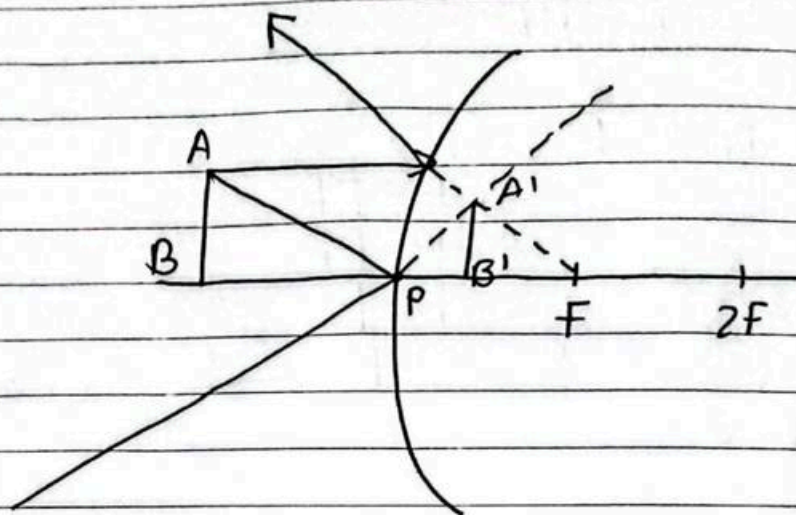
We know,

$$\begin{aligned} m &= \frac{\text{image size}}{\text{obj size}} \\ &= \frac{\text{image dis}}{\text{obj dis}} \end{aligned}$$

$$m = \frac{v}{u}$$



$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad \therefore \quad \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

1)
→

Here, When, an image AB is placed in front of a convex mirror then, A image A'B' is formed in the other side,
or

Nature of image,

Size = diminished
 Altitude = Erect
 location = diff side
 Type = Virtual,

→ It is used as rear-view mirror because this mirror forms a diminished and virtual image which widens the view needed for driving to see most of the are a. for

d)

→ Given,

-img- obj height = 5cm
image height = ?

$$u = 30\text{cm}$$

$$f = 20\text{cm}$$

We know,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{20} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{30 - 20}{20 \times 30}$$

$$v = \frac{60}{10}$$

$$v = 6\text{cm (as convex mirror)}$$

$$\frac{1}{v} = -0.08333333$$

$$v = -12\text{cm}$$

Now we know, magnification = $\frac{\text{image size}}{\text{obj size}}$

$$x = \frac{v}{u}$$

$$x = \frac{6}{30}$$

$$x = \frac{1}{5}$$

$$x = \frac{5}{30}$$

$$x = \frac{1}{6}$$

$$= 0.1667$$

Hence, the image is diminished, erect, and virtual.